

One student record, many controlled views.

Twenty operational modules plus a shared system layer, built as one product with one visual language — a desktop back office for Admin, Finance and Super Admin, and a mobile app for everyone who works in the building. Every colour, type step and layout rule below is fixed so no screen needs a bespoke decision.

MODULES	ROLES	SCREENS TOTAL	OF THOSE, MOBILE	TYPEFACE
20	9	123	71	Plus Jakarta Sans

PRINCIPLES

Six rules that settle most arguments

01 There is no student login

A student is a record, not an account. Every student-facing screen is designed as something a parent reads on behalf of a child they selected.

03 Marks and money need two people

Payroll is calculated by the system, reviewed by Finance and approved by Admin. Leave is requested by Faculty, approved by the Principal, and staffed by a section in-charge.

05 Status has one vocabulary

Approved, Pending and Overdue use the same chip, colour and wording in Attendance, Fees, Gate Pass and Transport alike.

02 Absence is the permission model

A module a role cannot use is removed from navigation, never greyed out. Dead UI advertises data the person is not allowed to know exists.

04 Assignment is not a role

Class Advisor, the three section in-charges and Community Head widen a Faculty account. They never produce a second login or a second design.

06 Colour is signal, not decor

Signal Blue marks the one primary action and every navigation tile. Green, amber and red are status only. Nothing is coloured to look nice.

Colour

One blue does all the work. It fills every circular tile, marks the active tab and carries the single primary action on a screen. Everything else is neutral so the tile grid stays the loudest thing on the page. Status colour never appears as a fill on anything a person can press.

Signal Blue #2B6FE0
PRIMARY
Every circular tile, the active tab, the one primary action on a screen, and the eyebrow on form field labels.

Deep Blue #1B3F9E
HEADER DEPTH
The far end of the header gradient and the active tab label. Never a fill on its own — it only appears under or beside Signal Blue.

NEUTRALS

App ground
#F4F6F9
Scrolling body

Surface
#FFFFFF
Card, sheet, row

Field
#F7F9FC
Input, counter tile

Hairline
#EAECF0
Card border, divider

Ink
#101828
Titles, values

Muted
#98A2B3
Eyebrows, timestamps

STATUS

Approved

Success
#027A48 on #ECFDF3
Present, approved, paid, on time, cleared, issued, published.

Pending

Pending
#B54708 on #FFFAEB
Awaiting approval, fee due, draft, open incident, under review.

Overdue

Critical
#B42318 on #FEF3F2
Absent, overdue, GPS offline, route deviation, escalated incident.

Typography

Plus Jakarta Sans, one family, four weights. The scale is short on purpose: a parent checking a fee and a warden marking night roll read the same seven steps. IBM Plex Mono appears only where a figure has to be compared down a column.

FAMILY	WEIGHTS	DATA FONT	NUMERALS	MEASURE CAP
Plus Jakarta Sans	500 • 600 • 700 • 800	IBM Plex Mono	tabular-nums, always	54ch
--eos-title	Bus Tracking		22 / 25 • 800	Header title
--eos-h1	Approval queue		28 / 34 • 700	Web page title
--eos-kpi	2,480		24 / 28 • 800	KPI, live counter
--eos-card	Madhavan B • U26CS214		15 / 20 • 800	Record card, sheet
--eos-body	Bus left the planned corridor near Peelamedu for 1.2 km.		13 / 19 • 400	Alert body, notes
--eos-tile	Rooms & Allocation		11.5 / 14 • 600	Under a tile
--eos-eyebrow	QUICK ACCESS		11 / 14 • 700 • .09em	Group header, meta key
--eos-num	₹ 18,400.00 • 94% • 412 / 480		15 / 20 • 500 mono	Marks, money, counts

Space, radius and depth

A 4pt rhythm on mobile, 8pt on web. Radius maps to component size, not taste.
Depth is one hairline plus one surface step; only the bottom sheet and desktop modal lift.

SPACING

space-1	4px	Icon-to-label, pill inner
space-2	8px	Chip row, button pair gap
space-3	12px	Row inner, avatar-to-text
space-4	14px	Card padding, field gap
space-5	18px	Body padding, sheet padding
space-6	22px	Between tile groups
space-8	32px	Between page sections
space-16	64px	Desktop page padding

RADIUS

	7px	Status pill, small button
	11px	Input, secondary button
	14px	Record card, list row
	16px	Section container
	22px	Bottom sheet top corners
	50%	Tile, avatar, chip

ELEVATION

- Flat** — cards, tiles, table rows. Hairline only.
- Raised** — primary button, circular tile. 0 4px 12px rgba(43,111,224,.25).
- Overlay** — bottom sheet and modal over a 45% ink scrim.

Shell

Two shells, one language. Mobile is a blue header, a scrolling body and a bottom bar of at most four tabs. Web is a fixed sidebar, a top bar and a workspace. The interface is never redesigned per role — permissions remove nav items rather than disabling them.

MOBILE · 390 × 844

Canvas	390 × 844 reference; 360 supported, 430 reflows
Header	Blue gradient, 22px bottom radius — back, title, subtitle, bell
Header height	96px with subtitle, 78px without
Body	18px side padding, 24px bottom, single scroll
Bottom bar	Two to four tabs, 23px glyph, 10.5px label, 22px indicator inset
Tile grid	Four columns, 6px column gap, 14px row gap
Touch target	44px minimum; 34px only for the three-up attendance pad

WEB · 1440

Canvas	1440 reference, 1280 content max
Top bar	56px — logo, search, notifications, profile
Sidebar	264px, collapses to a 64px icon rail below 1024
Nav order	Dashboard, Students, Parents, Faculty, Academics, Communities, Transport, Hostel, Finance, Reports, Settings
Page padding	64 / 40 / 16px by tier
Tables	Sticky header, pinned first column, right-aligned numerics
Below 768	Rail becomes a bottom tab bar, tables become cards

Permission rule. A module a role cannot use is never rendered greyed out. Absence is the permission model — dead UI hints at data the person is not allowed to know exists. A direct URL visit lands on the shared no-access state.

Component library

Twenty-five components carry all 123 screens. Every module is assembled from these — a new module never introduces a new control. There are five body patterns in total: tile grid, KPI feed, stat bars, record list and roster.

- 01 **Navigation tile** — 58px blue circle, 25px white glyph, label under. Four to a row.
- 02 **KPI card** — Eyebrow, value, one-line delta. Always four in a 2×2 grid.
- 03 **Record card** — Title, subtitle, status pill, two meta columns, optional action pair.
- 04 **Student profile card** — Photo, name, roll, class, guardian, day-scholar or hostel flag.
- 05 **Faculty card** — Name, subjects, classes, functional assignments, workload count.
- 06 **Roster row** — Initials disc, name, roll, then an attendance pad or a mark field.
- 07 **Attendance control** — Three fixed-width buttons — P, A, OD. Selected state is a fill.
- 08 **Mark input** — Mono, centred, numeric keypad, max-mark validated on blur.
- 09 **Status pill** — One of three status colours, 7px radius, never on a pressable surface.
- 10 **Filter chip row** — Horizontally scrolling, first chip active, counts inside the label.
- 11 **Search** — Sticky under the page title in every list module.
- 12 **Bottom sheet** — The only mobile modal. Handle, title, three to four fields, one action.
- 13 **Modal** — Web equivalent, 480px, same field rhythm as the sheet.
- 14 **Notification row** — Round icon, title, one sentence, relative timestamp.
- 15 **Timeline** — Approval chains, substitution history, gate-pass in and out.
- 16 **Progress bar** — Syllabus coverage, fee collection, hostel occupancy.
- 17 **Stat bar row** — Label, bar, right-aligned figure. Every comparison in the app.
- 18 **Data table** — Web only. Sticky header, bulk select for approvals, row actions.
- 19 **Map canvas** — Live bus markers, route-vs-actual overlay, status legend, playback.
- 20 **Calendar** — Academic calendar, exam schedule, community activity schedule.
- 21 **File attachment** — Dashed drop target, then a named row with size and remove.
- 22 **Confirmation dialog** — Names the value being committed and the actor committing it.
- 23 **Approval action** — Reject and Approve as a pair. Reject always demands a reason.
- 24 **Device status indicator** — Separates GPS offline from parked, and NFC unread from absent.
- 25 **Empty and error states** — Every list has no-data, no-permission and failed-load copy.

Roles and access

Nine logins. There is no student login — a student is a record their parent reads.

Class Advisor, the three section in-charges and Community Head are assignments layered onto a Faculty account, not additional roles. Creation is always separated from approval.

GROUP · WEB + MOBILE

Super Admin / Correspondent Read-only governance

Sees everything and commits almost nothing. School-wide dashboard, complete student and faculty records, finance, hostel, transport, sports and community overviews, plus the full audit trail of who changed what. Super Admin and Correspondent are the same role.

4 tabs · every module in view scope · full audit access

SCHOOL-WIDE · MOBILE

Principal / Vice Principal Mobile

One common login — there is no separate Vice Principal application, only a delegated scope on the same screens. Reads school attendance, performance and examination results, publishes announcements and SOPs, and approves faculty leave, which is the trigger for the substitution chain.

4 tabs · Vice Principal differs by row set, not by layout

BACK OFFICE · WEB

Admin Web

The configuration authority. Creates students, parents, faculty and every other account; sets academic years, standards, sections, subjects and the timetable; creates communities and appoints Community Heads; grants Class Advisor and the three section in-charge assignments.

11 sidebar items · creates every role except Admin itself

CLASSROOM · MOBILE

Faculty Mobile

One core role with assignment-based additions. Base access covers only assigned classes and subjects: mark attendance, set homework, upload LMS notes, create assessments and enter marks for own subject. Class Advisor widens this to the full record of one class; section in-charges gain substitution authority; Community Head gains one community.

4 tabs · conditional tiles appear only when an assignment is granted

FAMILY · MOBILE

Parent Mobile

Linked to one or more students, with a child switcher in the profile. Reads attendance, performance, homework, exams, timetable, LMS and mark sheets; pays fees; requests leave, OD and bonafide; sees hostel or transport depending on the child, and the community only if the child is a member.

4 tabs · every screen is scoped to the selected child

BACK OFFICE · WEB

Finance / Accounts Web

Money only — no academic or transport scope, and those modules are absent from the sidebar rather than disabled. Owns fee structures, concessions, payments, receipts, refunds, payroll and expenses, and generates the collection and outstanding reports the Principal reads.

6 sidebar items · reviews payroll, never gives final approval

RESIDENCE · MOBILE

Hostel Warden Mobile

Owens the building after hours. Room, floor and bed allocation, night roll call, visitor entry and exit, gate-pass approval with expected and actual return, complaints and incidents, and direct notification to the guardians of residents.

2 tabs · no academic or transport scope

FLEET · MOBILE

Transport Manager Mobile

Bus Tracking is the working surface, not a status page. Vehicles, GPS devices, routes and stop sequences, student allocation to bus, route and stop, driver and attendant records, maintenance, and acknowledgement of idle and deviation alerts.

3 tabs · the only role that receives deviation push alerts

GROUND · MOBILE

Sports Mobile

An independent login, not a Faculty assignment. Sports and categories, age groups, teams, captains and coaches, tournaments, fixtures, scores and rankings, equipment issue and return, and the achievement records parents read on the child profile.

3 tabs · writes to the student record, reads nothing else from it

Capability matrix

CAPABILITY	SUPER	PRIN/VP	ADMIN	FACULTY	PARENT	FINANCE	WARDEN	TRANSPORT
Mark class attendance	View	View	—	Own classes	Own child	—	—	—
Enter subject marks	View	View	—	Own subject	View	—	—	—
View all marks for a class	Yes	Yes	View	Advisor only	—	—	—	—
Create homework & LMS notes	View	View	—	Yes	View	—	—	—
Apply for leave	—	—	—	Yes	For child	—	—	—
Approve faculty leave	Override	Yes	—	—	—	—	—	—
Allocate substitution	Override	View	View	In-charge	—	—	—	—
Create students & accounts	Override	—	Yes	—	—	—	—	—
Configure academic structure	View	View	Yes	—	—	—	—	—
Create community	View	View	Yes	—	—	—	—	—
Manage a community	View	View	Assigns	Head only	View	—	—	—
Publish school announcement	Yes	Yes	Yes	—	—	—	—	—
Set fee structure & concessions	View	View	View	—	—	Yes	—	—
Pay fees	—	—	—	—	Yes	Records	—	—
Review payroll	Yes	—	—	—	—	Yes	—	—
Final payroll approval	Yes	—	Yes	—	—	—	—	—
Hostel rooms & night roll	View	View	Coord.	—	Own child	—	Yes	—
Approve gate pass	View	View	—	—	View	—	Yes	—
Manage routes & allocations	View	View	Coord.	—	—	—	—	Yes
Live bus location	All buses	All buses	View	Own bus	Child's bus	—	—	All buses
Deviation & idle alerts	Receives	—	—	—	—	—	—	Receives
Generate reports	View	View	Yes	—	—	Yes	Yes	Yes
Read the audit trail	Yes	—	Own scope	—	—	—	—	—

Records, not logins

Students, drivers, attendants, maintenance teams and buyers have no account. Admin creates the record; the owning role acts on their behalf.

Two-person money rule

Payroll is calculated by the system, reviewed by Finance and finally approved by Admin — who can never approve their own pay. Fee concessions follow the same split.

Alerts have one owner

Deviation and GPS alerts push to Transport Manager and Super Admin only. Everyone else reads the same map without a notification, so nobody assumes it was acknowledged.

Module map

Twenty modules, each with a stated intent, a platform, an owning set of roles and its full component inventory. Attendance flows into performance; assessment into results; fees into receipts; leave into substitution. Nothing else is coupled.

01 Dashboard

Answer "what needs me today" in five seconds, then get out of the way.

PLATFORM

Web + Mobile

SCREENS

4

ROLES

All 9 roles · scoped per role

COMPONENT INVENTORY

- Four KPI cards, chosen per role, with a one-line delta
- Alert feed: overdue passes, unmarked attendance, deviations, fee milestones
- Pending-action shortcuts that deep-link into the owning module
- Notification bell with unread dot and dropdown
- Child switcher on the Parent variant
- Last-updated stamp with pull to refresh

02 Identity, Roles & Assignments

Who can see and commit what — explicit, previewable, auditable.

PLATFORM

Web

SCREENS

5

ROLES

**Super Admin → Admin ·
Admin → all others**

COMPONENT INVENTORY

- User table: name, role, assigned scope, status
- Create user; Super Admin creates Admins, Admin creates everyone else
- Grant and revoke assignments: Class Advisor, Primary, Middle, 10–12, Community Head
- Role permission preview — exactly which tabs this account will get
- Deactivate, reset password, force sign-out
- Login activity log

03 Student Records

The single source of truth every other module points at.

PLATFORM

Web + Mobile

SCREENS

7

ROLES

**Admin owns · Principal,
Advisor and Parent read**

COMPONENT INVENTORY

- Student list with standard, section and status filters
- Create and edit student; standard, section, roll number, admission date
- Link parent and siblings; emergency contact
- Day scholar or hostel flag; transport allocation
- Document vault with expiry tracking
- Full profile: attendance, marks, homework, discipline, community, sports, hostel, transport, fees
- Student status: active, transferred, alumni

04 Parent & Guardian

One account, several children, one switch.

PLATFORM

Web + Mobile

SCREENS

4

ROLES

Admin creates · Parent maintains contact

COMPONENT INVENTORY

- Parent list with linked children count
- Create parent and link to one or more students
- Child switcher — every downstream screen re-scopes on selection
- Contact, emergency contact and communication preferences
- Faculty-parent case: one account carrying both experiences

05 Faculty & Staff

Assignments, workload and availability in one record.

PLATFORM

Web + Mobile

SCREENS

6

ROLES

Admin owns · Principal monitors

COMPONENT INVENTORY

- Faculty list with subject and class counts
- Create faculty; assign subjects and classes
- Functional assignment panel with effective dates
- Workload view: periods per week, free periods
- Faculty attendance and leave history
- Non-teaching staff records

06 Academic Configuration

Pre-primary to Grade 12 as configuration, never as code.

PLATFORM

Web

SCREENS

7

ROLES

Admin owns · Super Admin and Principal read

COMPONENT INVENTORY

- Academic year: label, term start and end, working days
- Standards from LKG to Grade 12, added and reordered freely
- Sections per standard with strength and advisor
- Subjects, subject groups and elective sets
- Terms and assessment types
- Examination types and grading scales
- Academic calendar with holidays and event types

07 Timetable & Substitution

Cover every period the day a teacher calls in.

PLATFORM

Web + Mobile

SCREENS

6

ROLES

Admin builds · Principal approves · in-charge staffs

COMPONENT INVENTORY

- Timetable grid builder with clash detection
- Per-faculty and per-section timetable views
- Leave request from the Faculty app
- Principal approve or reject with reason
- Section in-charge sees affected periods and available free faculty
- Assign substitute; timetable updates and both faculty are notified
- Substitution history per faculty and per section

08 Attendance

Four populations, four capture methods, one student record.

PLATFORM

Web + Mobile

SCREENS

6

ROLES

Faculty marks · Warden
night roll · NFC transport

COMPONENT INVENTORY

- Date and period selector; roster auto-loaded from the assignment
- Three-up pad per student: present, absent, on duty
- Mark all present, then correct the exceptions
- Live counters above the roster that recount on every tap
- Attendance history per student, per section, per subject
- Threshold alerting: sections below 85% surface on the Principal dashboard

09 Homework & LMS

Set the work, post the material, see who has not started.

PLATFORM

Web + Mobile

SCREENS

6

ROLES

Faculty creates · Parent
reads

COMPONENT INVENTORY

- Create homework: title, class, subject, due date, instructions, attachment
- Submission tracker with a submitted-of-total count
- Homework list filtered open, closed, all
- Upload notes and study material against a subject and unit
- Material list with type, size, posted date and author
- Parent view: due today, submitted, overdue

10 Assessment & Examination

From a class test to a published mark sheet, with one entry surface.

PLATFORM

Web + Mobile

SCREENS

8

ROLES

Faculty enters · in-charge
coordinates · Principal
reads

COMPONENT INVENTORY

- Create assessment: name, class, subject, max marks, date
- Mark entry roster — mono field per student, validated against max
- Live entered, pending and class-average counters
- Examination schedule: subject, date, time, room, seat
- Invigilator roster and hall readiness checklist
- Results publication with grade distribution
- Mark sheet per student, print-first
- Class comparison and subject standing

11 Communities

A club is a membership list with an owner, not a mailing list.

PLATFORM

Web + Mobile

SCREENS

6

ROLES

Admin creates · Faculty
head manages · Parent
reads

COMPONENT INVENTORY

- Create, edit and archive community
- Assign a Faculty account as Community Head
- Add and remove student members
- Activities with date, venue and participation record
- Community announcements — members and linked parents only
- Materials and achievement records

12 Finance & Fees

Every rupee has a head, a due date and a receipt.

PLATFORM

Web + Mobile

SCREENS

8

ROLES

Finance owns · Parent pays · Principal reads summary

COMPONENT INVENTORY

- Fee heads and structures by standard, day scholar or hosteller
- Student fee ledger with due dates and late fees
- Concessions and discounts through the approval hierarchy
- Online and offline payment capture
- Reconciliation queue for unmatched gateway settlements
- Receipt generation and refunds
- Collection, outstanding and daily-collection reports
- Parent view: payable, due date, pay, history

13 Payroll

Attendance becomes money only after two people have said yes.

PLATFORM

Web

SCREENS

7

ROLES

Finance reviews · Admin final approves

COMPONENT INVENTORY

- Pay period selector
- Salary structure: base, allowances, deductions
- Calculate from approved staff attendance
- Per-staff breakdown with remarks
- Review and send for approval
- Admin final approve or send back with reason
- Payslip generation and print, post-approval

14 Hostel

Know who is in the building, and who should be and is not.

PLATFORM

Mobile

SCREENS

8

ROLES

Warden owns · Parent reads own child

COMPONENT INVENTORY

- Blocks, floors, rooms and beds with occupancy and vacancy
- Allocation and allocation history
- Night roll call using the shared roster row
- Gate pass: request, approve, departure, expected and actual return
- Overdue return alerting to warden and guardian
- Visitor register with entry and exit
- Complaints and incidents with severity and escalation
- Direct notification to the guardians of residents

15 Transport & Live Tracking

Know where every bus is, and be told — not have to look — when one strays.

PLATFORM

Mobile

SCREENS

7

ROLES

Transport owns · Parent sees one bus

COMPONENT INVENTORY

- Vehicle register: number, capacity, GPS device, status
- Route builder with ordered stops and pickup and drop timings
- Student allocation to bus, route and stop
- Driver and attendant records with licence validity
- Live map with markers, legend and route-vs-actual overlay
- Deviation and idle alerts, acknowledged by the Transport Manager
- Maintenance schedule and cost by vehicle
- Parent view: one bus, ETA, last update, GPS state

16 NFC Transport Attendance

A tap at the door is the boarding record and the parent notification.

PLATFORM

Mobile + hardware

SCREENS

4

ROLES

System captures · Transport and Parent read

COMPONENT INVENTORY

- Card tap resolves to a unique student ID and verifies against the allocation
- Boarding and drop events with timestamp and stop
- Automatic parent notification on both events
- Bus-wise and route-wise attendance
- Missing-student view: allocated but not boarded
- Unread-card state distinguished from a genuine absence

17 Sports

Teams, fixtures and achievements that land on the student record.

PLATFORM

Mobile

SCREENS

8

ROLES

Sports login owns · Parent reads own child

COMPONENT INVENTORY

- Sports, categories, age groups and skill levels
- Student participation and performance records
- Team roster with captain and coach
- Tournament, fixture, score and ranking
- Equipment inventory, issue, return and damage
- Achievement entry with position, competition and certificate
- Achievement history written to the student profile
- Parent view: sports, teams, participation, results

18 Announcements & Notifications

One audience per message, chosen before it is written.

PLATFORM

Web + Mobile

SCREENS

5

ROLES

Principal, Admin, Warden,
Community Head

COMPONENT INVENTORY

- Compose with an explicit audience selector
- Audience scopes: school, standard, section, community, hostel, route
- Draft, schedule and publish
- Delivery and read counts per announcement
- Notification centre with unread state per role
- Attachment support

19 Reports & Analytics

Canned reports with one export path, no ad-hoc query builder.

PLATFORM

Web

SCREENS

5

ROLES

Admin, Finance, Warden
and Transport generate

COMPONENT INVENTORY

- Report type by module: attendance, results, collection, payroll, occupancy, trips
- Scope and period selector
- Summary totals with a chart-or-table toggle
- Comparison: period vs period, section vs section
- Export PDF, Excel and CSV
- Generated-report list with author and timestamp

20 Settings, Master Data & Audit

The lists every module depends on, and the record of who changed them.

PLATFORM

Web

SCREENS

6

ROLES

Super Admin owns ·
Admin school-level

COMPONENT INVENTORY

- School profile, branding and academic defaults
- Master data: document types, grade scales, fee heads, stop names
- System preferences: date format, currency, language
- Data export and backup
- Audit trail viewer: actor, action, entity, old value, new value, timestamp
- Audit filters by module, actor and date range

Build model — one app, every state designed

The whole feature surface is built once, as a single application. There is no Parent app and no Warden app — there is one screen register, and each screen is designed at full capability with its reduced states drawn alongside it.

01 One screen register

Every screen exists exactly once, at one route. Nothing is duplicated or redrawn per role.

02 Design the fullest version first

All columns, all filters, all actions. Reduced states are subtractions from that one layout, never a second design.

03 Three states per screen

Full, read-only, and hidden. A hidden screen visited directly lands on the shared no-access state.

04 Scope changes rows, not layout

One child, one section, one school: same columns, same controls. Only the row set and the scope selector differ.

How one screen serves every role

Attendance, as an example. One screen, one route, one layout — the action rail and the row set change, nothing else.

FULL

Mark and submit

The pad is live on every row; counters recount; submit is enabled.

READ-ONLY

Same roster, no pad

Marks render as values. History and counters still visible.

SCOPED

Own class • section • school

Same columns; the row set and the scope selector differ.

HIDDEN

Tab removed

Absent from the bottom bar. A direct route lands on no-access.

Core flows

Five chains carry the school. Each step names the role that acts and the state the record lands in.

Attendance to intervention

FACULTY · MOBILE Mark the period roster → submitted	SYSTEM Write to the student record → counted	SYSTEM Compare against the 85% threshold → flagged	ADVISOR See the student on the watch list → under review	PARENT Read the absence on the child profile → notified
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Assessment to mark sheet

FACULTY Create the assessment → open	FACULTY Enter marks against max → entered	IN-CHARGE Coordinate and close entry → locked	PRINCIPAL Publish results → published	PARENT Read the mark sheet → visible
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Leave to substitution

FACULTY Submit the leave request → pending	PRINCIPAL Approve or reject with reason → approved	SYSTEM Resolve date, period, class, subject → uncovered	IN-CHARGE Pick an available free faculty → assigned	SYSTEM Update the timetable, notify both → covered
--	--	---	---	--

Fee to receipt

FINANCE Publish the fee structure → active	SYSTEM Raise the student invoice → due	PARENT Pay from the Fees tile → processing	FINANCE Reconcile the settlement → matched	SYSTEM Issue the receipt, clear no-due → paid
--	--	--	--	---

Boarding to notification

STUDENT Tap the NFC card at the door → read	SYSTEM Verify against the allocation → boarded	SYSTEM Stamp bus, route, stop and time → logged	PARENT Receive the boarding push → notified	TRANSPORT Read the missing-student list → reconciled
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Notification and audience model

Every notification names one audience. Community announcements are the strict case: they reach members and their linked parents only, and a parent outside the community never learns it was sent.

SOURCE	AUDIENCE	EXAMPLE
School	All parents and staff	Parent-teacher meeting on Saturday 6 September.
Class	One section and its parents	CS-UG-C mentor hour moved to period 7.
Community	Members and their linked parents only	Coders Guild hackathon shortlist announced.
Transport	Parents of students on that route	Route 12 running 20 minutes late at Peelamedu.
NFC boarding	One guardian	Keerthi boarded Route 12 at Peelamedu, 07:42.
Hostel	Guardians of residents in scope	Expected return time passed for three students in Block A.
Finance	Parents with an open invoice	Term 2 fee of ₹18,400 payable before 15 September.
Examination	Parents of the examined cohort	CIA2 results published for Semester 1.
Faculty leave	Principal and the section in-charge	Leave request awaiting your approval.
Substitution	Both faculty and the section in-charge	You are covering CS101 period 2 on 4 September.
Deviation	Transport Manager and Super Admin	Route 12 left the planned corridor for 1.2 km.

Status and approval language

One state machine reused across every module. If a module needs a state that is not here, that is a system decision, not a module decision.

RECORD	CREATED BY	REVIEWED BY	COMMITTED BY
Class attendance	Faculty	—	Faculty submit
Subject marks	Faculty	Section in-charge	Principal publish
Faculty leave	Faculty	Principal / VP	Principal approve
Substitution	System (from leave)	Section in-charge	In-charge assign
Student leave / OD	Parent	Class Advisor	Advisor approve
Fee concession	Parent or Admin	Finance	Approval hierarchy
Payment	Parent or Finance	—	Finance reconcile
Payroll	System calculation	Finance	Admin final approve
Gate pass	Student or Parent	—	Warden approve
Bonafide request	Parent	Front office	Admin issue

Pending approval

Under review

Approved

Rejected

Due

Paid

Present

Absent

On duty

Overdue

Escalated

Published

Draft

Archived

Interaction states

Every interactive element ships six states. The attendance pad is the strictest case: a tap must be visible from arm's length, so the selected state is a filled swatch rather than a border change.

Selection is a fill, not a border

The P / A / OD pad and the filter chip both invert on selection. A teacher scanning 40 rows for the one they missed needs a solid block of colour, not a 1.5px edge change.

Loading is in place, not global

Rows fade to a skeleton bar; buttons keep their width and change their label. No full-page spinner — a stale roster with a loading strip beats a blank screen.

Transitions

120ms on colour and background, 180ms on sheet and drawer slide, 0ms on anything showing a number. Nothing animates a value the user is reading.

Focus is never removed

A 3px ring at 32% of the element's own colour. Mark entry on a web tablet is completable by keyboard alone, and the ring is how the user knows where they are.

Disabled always has a reason

A disabled primary carries a one-line note beneath it: "Mark every student before submitting". Never a dead button with no explanation.

Optimistic within a screen only

A tap on the pad updates the counter instantly. Nothing that leaves the device — submit, approve, pay — reports success before the server has confirmed it.

Forms and validation

Most creation in this product happens in a bottom sheet on a phone, often between periods. Validation is specific, in place, and phrased as what to do next.

RULE	APPLIES TO	EXAMPLE
Validate on blur, not per keystroke	All numeric fields	Nothing turns red while a teacher is still typing 8...8 into a mark field.
Marks are bounded by the assessment	Mark entry	"Above the maximum of 100 for this assessment. Check the paper total."
Required fields marked, not optional ones	Reject and override reasons	"Reason · required" in red above the control.
Reason is a pick list plus free text	Absence, rejection, override	Four common reasons as options, then "Something else...".
Never block on a plausible value	Attendance, fees, quantities	A whole section absent warns and records; only impossible values block.
One primary action per form	Every sheet and form	Submit, Approve or Pay — secondary actions never share its weight.
Unsaved work is never lost silently	Long forms, mobile entry	Leaving a part-marked roster keeps the draft and marks it in progress.

Numbers, units and copy

An ambiguous mark or an unclear fee costs a family money and a teacher an afternoon. These formats are fixed, not per-screen choices.

VALUE	RENDERED AS	RULE
Marks	88 / 100	Always against the maximum. Never a bare number, never a percentage unless the column header says so.
Percentage	94%	No decimals in a summary, one decimal only in a comparison table.
Money	₹ 18,400.00	Symbol before, space after it, two decimals in detail views.
Money in summary	₹ 4,42,000	Indian grouping, no decimals on totals — never abbreviate to 4.4L.
Counts	412 / 480	Present over total. The denominator is never dropped.
Student ID	U26CS211	Mono, never wrapped, never truncated in a list.
Date	31 Aug 2026	Never 08/31 or 31/08 — the month is always letters.
Time	14:20	24-hour, no am/pm, no seconds outside audit trails.
Relative time	12 s ago · 41 min ago	Only for live things: GPS pings, NFC taps, sync state. Never for records.
No value	—	An em dash in muted grey. Never 0, never blank, never "N/A".

Sentence case everywhere. Buttons, labels, headers and menu items. Only status chips and eyebrows use caps, and those are letter-spaced.

Alerts state the fact first. "3 gate passes overdue", then the sentence explaining it.

Name the actor. Approvals and overrides read "R. Subha · Class Advisor · 31 Aug 2026, 14:20". No passive voice on anything involving a child or money.

No exclamation marks. A warden reading an overdue pass at 21:30 does not need enthusiasm.

Buttons name the outcome. "Submit application", not "Submit". "Pay ₹18,400", not "Proceed".

Counts belong in the chip. "Pending (4)" — a user should never tap to learn a number.

Say what happened, then what to do. "Couldn't load the roster. Nothing was submitted — try again, or mark on paper and enter it later." Not "Something went wrong."

Use the words the school uses. Standard, section, period, roll number, class advisor, gate pass, day scholar. Never "asset" for a bus or "user" for a child.

Icons, charts and imagery

Restrained on purpose. Nine roles share one icon set, and the circular tile is the only place an icon carries weight.

One icon set, 24px, 1.5px stroke

Outline only, currentColor, drawn on a 24px box. Rendered white inside the circular tile and at 38% opacity on an inactive tab. No icon appears in navigation without a label.

Three chart types only

Horizontal bar for comparison, stacked bar for grade mix, line for a trend over time. No pie charts and no line graphs on a 390px screen.

Photography is identity or evidence

Student and staff portraits, and user-captured incident photos. No stock imagery of classrooms anywhere in the product.

No emoji, ever

Not in copy, not in chips, not in empty states. They break at small sizes on cheap Android hardware and read as unserious beside a fee figure or a discipline record.

Charts use one colour

Signal Blue for the series, muted grey for the track. Status colours keep their meaning inside charts — red is never just "series 3".

Mark sheets and receipts are print-first

Mono figures, no colour dependency, readable on a mono laser. Colour carries no meaning on either document.

Accessibility and conditions of use

The hardest environment this product runs in is a warden's phone at 21:30 in a corridor, and a parent's five-year-old Android on a bus stop pavement. That sets the floor for everything.

Contrast floors

Body text on white is 12.1:1. Nothing carrying meaning falls below 4.5:1, and no text sits lighter than #98A2B3 on white — which is why muted grey never carries a value.

Touch targets

44px minimum, with 8px between adjacent targets. The three-up attendance pad is the one exception at 34px tall, and it is compensated by full-width row spacing.

Full keyboard path on web

Mark entry and payroll approval are completable without a mouse: tab order follows visual order, Enter commits the primary action, Esc closes the drawer.

Colour is never the only signal

Every status chip carries a word as well as a colour. Attendance shows a letter, not just a coloured square. The design survives being printed in mono.

Text scales to 200%

No fixed-height text containers and no truncation of a student name or a status word. Tables reflow to cards rather than clipping.

Two languages, one layout

Regional-language and English labels are budgeted for from the start — every label box tolerates 40% more characters without breaking its row.

Live examples

Part of the register is built and interactive. These are the reference for anything drawn next.

MOBILE · 390 × 844

Role-switching app shell

All nine roles with working tabs, tile grids, live attendance and mark entry, create sheets and the bus map.

DOCUMENT

Role and access matrix

Every tab and tile per role with its access verb, plus the shared-tile comparison table.

LIBRARY

Design system specimens

Colour, type scale, spacing, radius, and rendered specimens of every component in section 05.